

Benchmark results compiled with -O2: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lisstt	handrolled.lisstt
countdown.shallow.10000	172.066	154.12	1209.482	4.911	142.74	281.149	257.721	2.494	86.49	186.394	1100.91	4.951	4.933	-	-	-
countdown.deep.10000	1286.715	571.096	2309.474	2683.87	146.023	279.041	162.491	2.494	168.262	191.453	3509.802	4.971	4.954	65.201	-	-
countdown.very-deep.10000	9251.606	5075.962	11114.513	-	162.11	276.788	164.138	2.497	1314.537	490.748	23080.338	4.98	4.939	65.032	-	-
catch.shallow.10000	269.634	35.393	1117.336	20.195	1065.504	400322.2	245.976	16.077	73.806	835.352	834.281	20.257	20.36	-	-	-
catch.deep.10000	3259.689	34.398	6536.075	1700.774	1077.138	399330.6	174.191	16.057	240.985	846.729	851.639	20.353	20.243	167.612	-	-
catch.very-deep.10000	40653.7	32.4	40521.762	-	1201.884	378063.8	175.966	15.978	1784.356	857.593	849.1	20.287	20.39	165.591	-	-
local.shallow.10000	918.012	616.257	2193.356	4.974	536.764	398140.6	198.672	2.5	134.544	781.972	782.685	4.982	4.974	-	-	-
local.deep.10000	3847.183	454.187	5099.9	4.971	537.659	384135.2	202.792	2.498	218.587	790.822	784.5	4.98	4.972	201.48	-	-
local.very-deep.10000	36030.113	456.704	28175.638	-	570.613	382941.1	203.396	2.497	896.762	789.211	797.438	3537.939	3524.177	201.966	-	-
nondet.shallow.32	2006.941	3378.8	8074.909	240.779	-	24556.725	756.505	79.51	898.865	1716.709	1725.028	225.101	241.682	-	347.535	347.748
nondet.deep.32	5511.312	5710.284	8071.544	2404.49	-	50724.2	764.697	79.321	603.822	5635.622	5650.906	242.35	241.867	990.087	62547.275	62482.6
nondet.very-deep.32	41445.35	25597.575	7755.594	-	-	392451.4	770.11	79.857	614.329	38921.3	38913.15	248.897	248.269	988.955	667690.8	667393.6

Benchmark results compiled with -O2: mean time (% of fastest)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lisstt	handrolled.lisstt
countdown.shallow.10000	6899.1%	6179.5%	48494.9%	196.9%	5723.3%	11272.8%	10333.5%	100.0%	3467.9%	7473.6%	44141.6%	198.5%	197.8%	-	-	-
countdown.deep.10000	51596.5%	22900.6%	92608.6%	107621.7%	5855.4%	11189.4%	6515.8%	100.0%	6747.2%	7677.2%	140741.0%	199.4%	198.7%	2614.5%	-	-
countdown.very-deep.10000	370462.6%	203257.0%	445050.0%	-	6491.4%	11083.4%	6572.6%	100.0%	52638.1%	19651.1%	924207.2%	199.4%	197.8%	2604.1%	-	-
catch.shallow.10000	1677.2%	220.2%	6950.0%	125.6%	6627.6%	2490052.0%	1530.0%	100.0%	459.1%	5196.0%	5189.3%	126.0%	126.6%	-	-	-
catch.deep.10000	20300.6%	214.2%	40705.3%	10592.1%	6708.2%	2486944.1%	1084.8%	100.0%	1500.8%	5273.2%	5303.8%	126.8%	126.1%	1043.9%	-	-
catch.very-deep.10000	254439.7%	202.8%	253614.0%	-	7522.2%	2366191.6%	1101.3%	100.0%	11167.8%	5367.4%	5314.3%	127.0%	127.6%	1036.4%	-	-
local.shallow.10000	36720.7%	24650.4%	87734.8%	198.9%	21470.7%	15925725.9%	7946.9%	100.0%	5381.8%	31279.1%	31307.6%	199.3%	199.0%	-	-	-
local.deep.10000	154029.1%	18184.2%	204184.0%	199.0%	21526.2%	15379569.5%	8119.1%	100.0%	8751.5%	31662.0%	31408.9%	199.4%	199.0%	8066.6%	-	-
local.very-deep.10000	1443092.1%	18292.1%	1128501.6%	-	22854.4%	15337705.7%	8146.5%	100.0%	35917.5%	31609.8%	31939.3%	141702.9%	141151.7%	8089.2%	-	-
nondet.shallow.32	2524.1%	4249.5%	10155.8%	302.8%	-	30885.0%	951.5%	100.0%	1130.5%	2159.1%	2169.6%	283.1%	304.0%	-	437.1%	437.4%
nondet.deep.32	6948.2%	7199.0%	10175.9%	3031.4%	-	63948.4%	964.1%	100.0%	761.2%	7104.9%	7124.1%	305.5%	304.9%	1248.2%	78853.8%	78772.3%
nondet.very-deep.32	51899.2%	32054.1%	9711.8%	-	-	491440.7%	964.4%	100.0%	769.3%	48738.5%	48728.3%	311.7%	310.9%	1238.4%	836104.7%	835732.5%

Benchmark results compiled with -O2: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lisstt	handrolled.lisstt
countdown.shallow.10000	1760162	1600145	13361481	24	1121570	2087371	2480242	24	1040121	3441170	14241121	24	24	-	-	-
countdown.deep.10000	17760483	10400173	37363380	27121286	1136505	2088075	2000194	24	3920274	3442691	41443963	24	24	1200100	-	-
countdown.very-deep.10000	161763304	89600730	253380957	-	1410353	2085107	2000226	104	20401224	3776413	286589490	104	104	1200132	-	-
catch.shallow.10000	4069612	65596	7764897	65586	7846353	413467414	1687795	65585	723953	6081394	6081394	65586	65586	-	-	-
catch.deep.10000	16870482	66058	55153744	16165107	7864171	414305494	1527786	65585	3367875	6083955	6083954	65586	65585	1527786	-	-
catch.very-deep.10000	132077892	70018	371972753	-	8163212	421556572	1527777	65585	18074259	6139778	6139778	65586	65586	1527787	-	-
local.shallow.10000	7205058	4793294	12630078	24	4326220	409641299	1669450	24	1029434	5513578	5513578	24	24	-	-	-
local.deep.10000	16006054	4793732	30232297	24	4314334	410402734	1589474	24	2549554	5515826	5515826	64	64	1589474	-	-
local.very-deep.10000	95215313	4797716	188651339	-	4648661	417691622	1589474	64	9830346	5535987	5535986	28886124	28886111	1589475	-	-
nondet.shallow.32	13454124	15889970	43492626	36069007	-	111261132	1531330	108345	1532765	22646075	22646075	1987330	2003987	-	3664108	3664101
nondet.deep.32	67560404	45646135	43569971	26810555	-	383227083	1531411	108345	1535162	68636524	68636530	1969098	2010674	7408482	469819536	469819525
nondet.very-deep.32	554516648	313450797	44266212	-	-	4657484945	1531410	108345	1559642	483091545	483091454	2035339	2076914	7408482	5480086293	5480085304

Benchmark results compiled with -O2: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lisstt	handrolled.lisstt
countdown.shallow.10000	7334008.3%	6667270.8%	55672837.5%	100.0%	4673208.3%	8697379.2%	10334341.7%	100.0%	4333837.5%	14338298.3%	59338004.2%	100.0%	100.0%	-	-	-
countdown.deep.10000	74002012.5%	43334054.2%	155680750.0%	113005358.3%	4735437.5%	8700312.5%	8334141.7%	100.0%	16334475.0%	14344545.8%	172683179.2%	100.0%	100.0%	5000416.7%	-	-
countdown.very-deep.10000	155541638.5%	86154548.1%	243635335.6%	-	1356108.7%	2004910.6%	1923294.2%	100.0%	19616580.8%	3631166.3%	275566817.3%	100.0%	100.0%	1153973.1%	-	-
catch.shallow.10000	6205.1%	100.0%	11839.4%	100.0%	11963.6%	630429.8%	2573.4%	100.0%	1103.8%	9272.5%	9272.5%	100.0%	100.0%	-	-	-
catch.deep.10000	25723.1%	100.7%	84095.1%	24647.6%	11990.8%	631707.7%	2329.5%	100.0%	5135.1%	9276.4%	9276.4%	100.0%	100.0%	2329.5%	-	-
catch.very-deep.10000	201384.3%	106.8%	567161.3%	-	12446.8%	642763.7%	2329.5%	100.0%	27558.5%	9361.6%	9361.6%	100.0%	100.0%	2329.5%	-	-
local.shallow.10000	30021075.0%	19972058.3%	52825325.0%	100.0%	18025916.7%	1708838745.8%	695041.7%	100.0%	4289308.3%	22973241.7%	22973241.7%	100.0%	100.0%	-	-	-
local.deep.10000	66601891.7%	19973883.3%	125967904.2%	100.0%	18088891.7%	1710011391.7%	6622808.3%	100.0%	10623141.7%	22982608.3%	22982608.3%	266.7%	266.7%	6622808.3%	-	-
local.very-deep.10000	148773926.6%	7496431.2%	294767717.2%	-	7263532.8%	65264319.4%	2483553.1%	100.0%	15359915.6%	8649979.7%	8649978.1%	45134568.8%	45134548.4%	2483554.7%	-	-
nondet.shallow.32	12417.9%	14666.1%	40142.7%	3329.1%	-	102691.5%	1413.4%	100.0%	1414.7%	20901.8%	20901.8%	1834.3%	1849.6%	-	3381.9%	3381.9%
nondet.deep.32	62356.7%	42130.4%	40214.1%	24745.5%	-	353710.0%	1413.5%	100.0%	1416.9%	63350.0%	63350.0%	1817.4%	1855.8%	6837.9%	433632.9%	433632.9%
nondet.very-deep.32	511806.4%	289308.0%	40856.7%	-	-	4298753.9%	1413.5%	100.0%	1439.5%	445882.6%	445882.6%	1878.6%	1916.9%	6837.9%	5057996.5%	5057995.6%

Benchmark results compiled with -O0: mean time (microseconds)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.lstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lstt	handrolled.lstt
countdown.shallow.10000	792.759	516.948	1748.041	1703.557	722.33	866.258	1687.371	106.123	1685.148	856.807	2200.959	669.582	565.993	—	—	—
countdown.deep.10000	2967.139	974.746	2846.219	10662.95	727.26	867.081	1705.43	106.175	3345.491	1475.984	5280.878	5208.206	5917.894	1725.277	—	—
countdown.very-deep.10000	20043.175	5312.859	11730.944	82674.1	800.642	910.969	1737.155	106.078	16337.806	6342.8	28817.138	43032.475	48549.575	1713.989	—	—
catch.shallow.10000	1091.311	303.176	2907.922	3598.633	2122.874	371472.4	954.964	352.473	1070.519	1689.797	1695.923	253.782	222.551	—	—	—
catch.deep.10000	7926.322	309.3	7003.394	15848.487	2144.468	368451.6	1049.974	345.521	3094.384	1676.444	1688.159	4129.344	4315.866	1135.516	—	—
catch.very-deep.10000	77454.5	309.35	40557.675	110292.65	2294.166	384279.8	1072.698	345.133	16576.531	1665.527	1664.378	39229.814	45862.85	1154.688	—	—
local.shallow.10000	4766.106	669.727	2718.63	1434.17	1060.92	399788.4	637.746	298.428	666.043	1396.529	1401.706	351.334	332.067	—	—	—
local.deep.10000	13609.319	673.54	5695.106	6022.478	1075.556	384788.7	640.509	298.22	1648.311	1410.093	1407.305	1813.544	1638.51	646.902	—	—
local.very-deep.10000	82149.4	677.427	29364.475	38495.825	1148.224	383264.9	650.584	298.846	8663.422	1424.138	1418.992	12243.737	11146.062	657.346	—	—
nondet.shallow.32	7710.5	5946.584	13892.144	2234.161	—	28134.325	4197.914	5849.553	4307.191	404.188	3072.256	2523.436	2014.584	—	7542.253	7003.05
nondet.deep.32	16054.781	8000.603	23349.45	10812.862	—	54159.425	4333.808	5888.93	9388.269	7886.603	7831.141	6965.269	7244.153	4588.006	81381.15	81821.65
nondet.very-deep.32	86959.85	28236.275	99687	110878.85	—	392578.2	4338.966	5848.25	54557.275	47757.875	48046.225	50986.425	54887.45	4622.903	737098.2	743571.1

Benchmark results compiled with -O0: allocated bytes

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lstt	handrolled.lstt
countdown.shallow.10000	10640948	6080478	17281886	11920971	5202083	6268900	10801445	400153	11601229	7041771	19361627	4080275	2880389	—	—	—
countdown.deep.10000	38643779	14881349	41284588	63365432	5218174	6269552	11213180	400153	15371645	9444841	48965635	30882914	30882667	11365142	—	—
countdown.very-deep.10000	290665289	94088347	257307421	524203694	5500291	6267135	11319006	400153	44261452	31071203	315398002	272104882	282905446	11398614	—	—
catch.shallow.10000	7914022	2571789	22351561	13075339	12013517	415982702	5434003	1462349	6393818	8161829	8161832	1673347	1593377	—	—	—
catch.deep.10000	33116444	2572605	57554635	64806353	12032293	416790307	5845722	1462350	10164223	8165128	8164621	16926443	18046823	5837666	—	—
catch.very-deep.10000	297290493	2579445	374381093	598207170	12340561	424127595	5951270	1462349	39053660	8227059	8221509	177239336	213164916	5871213	—	—
local.shallow.10000	14981955	5771851	14008722	7688268	6726770	411563591	4008483	1767861	4648303	7193877	7193877	2327813	2327869	—	—	—
local.deep.10000	34185134	5772067	31611570	28490777	6742834	412324679	4020203	1767861	6818582	7197165	7196661	9128590	9048648	4012086	—	—
local.very-deep.10000	227160792	5779307	190035656	215712384	7058083	419623357	4125675	1767861	21307005	7226325	7220781	70335380	66655440	4045555	—	—
nondet.shallow.32	53827237	34532261	81644041	18238563	—	131921113	24814522	53824727	296833900	30882032	30882027	17397027	12390869	—	74123186	69116717
nondet.deep.32	137695708	64288505	200681833	63405695	—	403888242	26178325	53824725	42408356	80929300	80928819	43632237	39124643	28604571	543616138	539108949
nondet.very-deep.32	892512168	332096082	1272018001	469909925	—	4678155598	26284095	53824723	139881902	531355718	531350368	279746154	280236073	28636306	4837187723	4837677497

Benchmark results compiled with -O0: allocated bytes (% of least allocated)

Test case	heftia	freer	polysemy	fused	effectful	eff	effective	effective.fstg	effective.naive	mp	mp.safe	mtl.logict	handrolled.logict	effective.lstg	mtl.lstt	handrolled.lstt
countdown.shallow.10000	2659.2%	1519.5%	4318.8%	2979.1%	1300.0%	1566.6%	2699.3%	100.0%	2899.2%	1759.8%	4838.6%	1019.7%	719.8%	—	—	—
countdown.deep.10000	9657.3%	3718.9%	10317.2%	15835.3%	1304.0%	1566.8%	2802.2%	100.0%	3841.4%	2360.3%	12236.7%	7717.8%	7717.7%	2840.2%	—	—
countdown.very-deep.10000	72638.5%	23513.1%	64302.3%	131000.8%	1374.5%	1566.2%	2828.7%	100.0%	11061.1%	7764.8%	78819.4%	68000.2%	70699.3%	2848.6%	—	—
catch.shallow.10000	541.2%	175.9%	1528.5%	894.1%	821.5%	28446.2%	371.6%	100.0%	437.2%	558.1%	558.1%	114.4%	109.0%	—	—	—
catch.deep.10000	2264.6%	175.9%	3935.8%	4431.7%	822.8%	28501.4%	399.7%	100.0%	695.1%	558.4%	558.3%	1157.5%	1234.1%	399.2%	—	—
catch.very-deep.10000	20329.7%	176.4%	25601.4%	40907.3%	843.9%	29003.2%	407.0%	100.0%	2670.6%	562.6%	562.2%	12120.2%	14576.9%	401.5%	—	—
local.shallow.10000	847.5%	326.5%	792.4%	434.9%	380.5%	23280.3%	226.7%	100.0%	262.9%	406.9%	406.9%	131.7%	131.7%	—	—	—
local.deep.10000	1933.7%	326.5%	1788.1%	1611.6%	381.4%	23323.4%	227.4%	100.0%	385.7%	407.1%	407.1%	516.4%	511.8%	226.9%	—	—
local.very-deep.10000	12849.5%	326.9%	10749.5%	12201.9%	399.2%	23736.2%	233.4%	100.0%	1205.3%	408.8%	408.4%	3978.6%	3770.4%	228.8%	—	—
nondet.shallow.32	434.4%	278.7%	658.9%	147.2%	—	1064.7%	200.3%	434.4%	239.6%	249.2%	249.2%	140.4%	100.0%	—	598.2%	557.8%
nondet.deep.32	526.0%	245.6%	766.6%	242.2%	—	1542.8%	100.0%	205.6%	162.0%	309.1%	309.1%	166.7%	149.5%	109.3%	2076.6%	2059.4%
nondet.very-deep.32	3395.6%	1263.5%	4839.5%	1787.8%	—	17798.4%	100.0%	204.8%	532.2%	2021.6%	2021.6%	1064.3%	1066.2%	108.9%	18403.5%	18405.3%